Exponentials & Logarithms

Evaluating exponential expressions

1 Evaluate 2^5.

2 Evaluate 3^{-2}.

3 Evaluate $16^{1/2}$.

4 Evaluate $27^{2/3}$.

Evaluating logarithms

5 Evaluate $\log_2 32$.

6 Evaluate $\log_3 81$.

7 Evaluate $\log_5 1$.

8 Evaluate $\log_{10} 0.01$.

Log laws: product and quotient

9 Write $\log_2(8x)$ as a sum of logarithms.

10 Write $\log_3\left(\frac{x}{9}\right)$ as a difference of logarithms.

11 Combine $\log_5 5 + \log_5 25$ into a single logarithm and evaluate.

12 Combine $\log_2 24 - \log_2 3$ into a single logarithm and evaluate.

Log laws: the power rule

13 Write $\log(x^5)$ using the power rule.

14 Evaluate $\log_2(4^3)$ using the power rule.

15 Write $2\log_3 x + \log_3 y$ as a single logarithm.

Expanding and condensing logarithmic expressions

16 Expand $\log\left(\frac{x^3y}{z^2}\right)$ fully.

17 Condense $3\ln x - 2\ln y$ into a single logarithm.

18 Expand $\log_2\left(\sqrt{x}\cdot y^3\right)$ fully.

Solving exponential equations (matching bases)

19 Solve $2^x = 32$.

20 Solve $3^{x+1} = 81$.

21 Solve $5^{2x} = 125$.

Solving exponential equations using logarithms

22 Solve $2^x = 10$ for x, rounded to three decimal places.

23 Solve $5^x = 40$ for x, rounded to three decimal places.

24 Solve $3^{2x} = 50$ for x, rounded to three decimal places.

25 Solve $e^x = 15$ for x, rounded to three decimal places.

Solving logarithmic equations

26 Solve $\log_2(x) = 5$.

27 Solve $\log_3(x-1) = 2$.

28 Solve $\log(x) + \log(x-3) = 1$.

29 Solve $\log_2(x+3) + \log_2(x-1) = 5$.

Natural logarithms and e

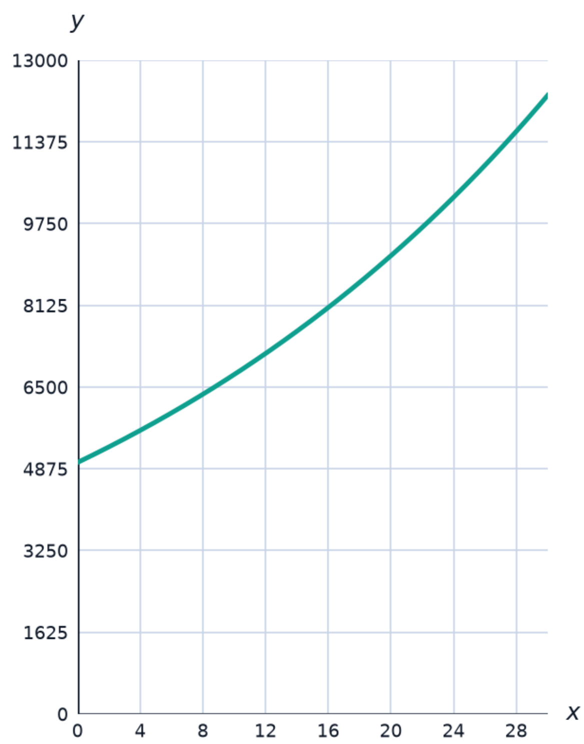
30 Evaluate $\ln(e^3)$.

31 Evaluate $\ln(1)$.

32 Solve $\ln(x) = 2$ for x, in terms of e.

Exponential growth and decay models

33 The population of a town is modeled by $P(t) = 5000e^{0.03t}$, where t is in years. Find $P(10)$, rounded to the nearest whole number.



34 A radioactive substance decays according to $A(t) = 200e^{-0.05t}$ (grams). Find $A(20)$, rounded to one decimal place.

35 Determine whether $N(t) = 50e^{0.02t}$ models growth or decay, and justify your answer.

Word problems: exponential models in context

36 This question has two parts. An investment of 2000 dollars grows according to $A(t) = 2000e^{0.04t}$, where t is in years.

a Find the value of the investment after 5 years, rounded to the nearest dollar.

b Find how many years it takes for the investment to double, rounded to one decimal place (using $\ln 2 \approx 0.693$).

37 This question has two parts. A cup of coffee cools according to $T(t) = 20 + 60e^{-0.1t}$ (°C), where t is minutes after brewing and 20°C is room temperature.

a Find the temperature after 15 minutes, rounded to one decimal place (using $e^{-1.5} \approx 0.2231$).

b Find how many minutes it takes for the temperature to drop to 35°C, rounded to one decimal place (using $\ln 4 \approx 1.386$).

MCQ — Evaluating exponential expressions

38 Evaluate 3^4.

a. 81

b. 243

c. 12

d. 64

39 Evaluate 4^{-2}.

a. $-\frac{1}{16}$

b. $\frac{1}{16}$

c. $\frac{1}{8}$

d. 16

40 Evaluate $25^{1/2}$.

a. 5

b. 50

c. 12.5

d. 625

41 Evaluate $8^{2/3}$.

a. 5.33

b. 4

c. 2

d. 16

42 Evaluate 2^{-3}.

a. -8

b. $\frac{1}{6}$

c. 8

d. $\frac{1}{8}$

43 Evaluate $32^{3/5}$.

a. 2

b. 16

c. 8

d. 32

MCQ — Evaluating logarithms

44 Evaluate $\log_3 27$.

a. 3

b. 1

c. 27

d. 9

45 Evaluate $\log_2 64$.

a. 32

b. 8

c. 5

d. 6

46 Evaluate $\log_7 1$.

a. 1

b. Undefined

c. 7

d. 0

47 Evaluate $\log_{10} 0.001$.

a. 0.001

b. -3

c. -2

d. 3

48 Evaluate $\log_4 16$.

a. 4

b. 8

c. 2

d. 16

49 Evaluate $\log_5 125$.

a. 5

b. 3

c. 1

d. 25

MCQ — Log laws: product and quotient

50 Write $\log_3 (9x)$ as a sum of logarithms.

a. $2 + \log_3 x$

b. $2\log_3 x$

c. $9 + \log_3 x$

d. $\log_3 9 + \log_3 x$ only

51 Write $\log_5 \left(\frac{x}{25}\right)$ as a difference of logarithms.

a. $\log_5 x - 25$

b. $2 - \log_5 x$

c. $\log_5 x - 2$

d. $\frac{\log_5 x}{2}$

52 Combine $\log_2 4 + \log_2 8$ into a single logarithm and evaluate.

a. 32

b. 5

c. 12

d. 2

53 Combine $\log_3 54 - \log_3 2$ into a single logarithm and evaluate.

a. 27

b. 3

c. 26

d. 52

MCQ — Log laws: the power rule

54 Write $\log(x^7)$ using the power rule.

a. $7\log x$

b. $x^7\log x$

c. $\log x^7$ unchanged

d. $\log(7x)$

55 Evaluate $\log_3(9^2)$ using the power rule.

a. 18

b. 81

c. 2

d. 4

56 Write $3\log_2 x + \log_2 y$ as a single logarithm.

a. $3\log_2(xy)$

b. $\log_2(x^3 + y)$

c. $\log_2(x^3 y)$

d. $\log_2(3xy)$

57 Write $4\log x - \log y$ as a single logarithm.

a. $\log\left(\frac{x^4}{y}\right)$

b. $\log(4x - y)$

c. $4\log\left(\frac{x}{y}\right)$

d. $\log\left(\frac{4x}{y}\right)$

MCQ — Expanding and condensing logarithmic expressions

58 Expand $\log\left(\frac{x^2 y}{z^3}\right)$ fully.

a. $2\log x + \log y - 3\log z$

b. $2\log x - \log y - 3\log z$

c. $\log x^2 + \log y - \log z^3$ unchanged

d. $2\log x + \log y + 3\log z$

59 Condense $2\ln x - 3\ln y$ into a single logarithm.

a. $\ln\left(\frac{x^2}{y^3}\right)$

b. $\ln\left(\frac{x^2}{y}\right) - 3$

c. $\ln\left(\frac{x^3}{y^2}\right)$

d. $\ln(x^2 - y^3)$

60 Expand $\log_2\left(\sqrt{x}\cdot y^4\right)$ fully.

a. $\frac{1}{2}\log_2 x + 4\log_2 y$

b. $\frac{1}{2}\log_2 x + \log_2 y^4$ unchanged

c. $2\log_2 x + 4\log_2 y$

d. $\frac{1}{4}\log_2 x + 2\log_2 y$

61 Condense $\frac{1}{2}\log x + 2\log y$ into a single logarithm.

a. $\log(x\cdot y^2)$

b. $\log\left(\frac{\sqrt{x}}{y^2}\right)$

c. $\log(\sqrt{x}\cdot y^2)$

d. $\log(2\sqrt{x}\cdot y)$

MCQ — Solving exponential equations (matching bases)

62 Solve $2^x = 64$.

a. $x = 8$

b. $x = 6$

c. $x = 32$

d. $x = 5$

63 Solve $3^{x+2} = 27$.

a. $x = 9$

b. $x = 1$

c. $x = 5$

d. $x = 3$

64 Solve $5^{3x} = 125$.

a. $x = \frac{1}{3}$

b. $x = 41.67$

c. $x = 3$

d. $x = 1$

65 Solve $4^{x} = \frac{1}{16}$.

a. $x = 4$

b. $x = -4$

c. $x = -2$

d. $x = 2$

66 Solve $2^{2x-1} = 32$.

a. $x = 3$

b. $x = 2.5$

c. $x = 6$

d. $x = 5.5$

MCQ — Solving exponential equations using logarithms

67 Solve $2^{x} = 12$ for x, rounded to three decimal places.

a. 3.585

b. 3.170

c. 4.000

d. 2.485

68 Solve $7^x = 50$ for x, rounded to three decimal places.

- **a.** 7.143
- **b.** 2.010
- **c.** 3.000
- **d.** 1.505

69 Solve $4^{2x} = 90$ for x, rounded to three decimal places.

- **a.** 6.492
- **b.** 3.246
- **c.** 1.623
- **d.** 0.811

70 Solve $e^x = 25$ for x, rounded to three decimal places.

- **a.** 25.000
- **b.** 2.303
- **c.** 3.219
- **d.** 12.500

MCQ — Solving logarithmic equations

71 Solve $\log_3(x) = 4$.

- **a.** $x = 7$
- **b.** $x = 12$
- **c.** $x = 64$
- **d.** $x = 81$

72 Solve $\log_2(x-1)=3$.

a. $x=6$

b. $x=9$

c. $x=7$

d. $x=8$

73 Solve $\log(x)+\log(x-3)=1$.

a. $x=-2$

b. $x=5$

c. $x=10$

d. $x=13$

74 Solve $\log_2(x+2)+\log_2(x-2)=5$.

a. $x=6$

b. $x=-6$

c. $x=34$

d. $x=18$

MCQ — Natural logarithms and e

75 Evaluate $\ln(e^5)$.

a. 0

b. e^5

c. 1

d. 5

76 Evaluate $\ln\left(\frac{1}{e}\right)$.

a. 1

b. e

c. -1

d. 0

77 Solve $\ln(x) = 3$ for x, in terms of e.

a. $x = e^{1/3}$

b. $x = 3$

c. $x = e^3$

d. $x = 3e$

78 Evaluate $e^{\ln 7}$.

a. 7

b. $\ln 7$

c. 1

d. e^7

MCQ — Exponential growth and decay models

79 The population of a town is modeled by $P(t) = 8{,}000e^{0.025t}$, where t is in years. Find $P(15)$, rounded to the nearest whole number.

a. $8{,}300$

b. $11{,}000$

c. $11{,}640$

d. $12{,}000$

80 A radioactive substance decays according to $A(t) = 150e^{-0.04t}$ (grams). Find $A(25)$, rounded to one decimal place.

a. 60.7

b. 50.0

c. 75.0

d. 55.2

81 Determine whether $N(t) = 80e^{-0.015t}$ models growth or decay.

a. Decay

b. Neither

c. Growth

d. Cannot be determined

82 Determine whether $M(t) = 30e^{0.01t}$ models growth or decay.

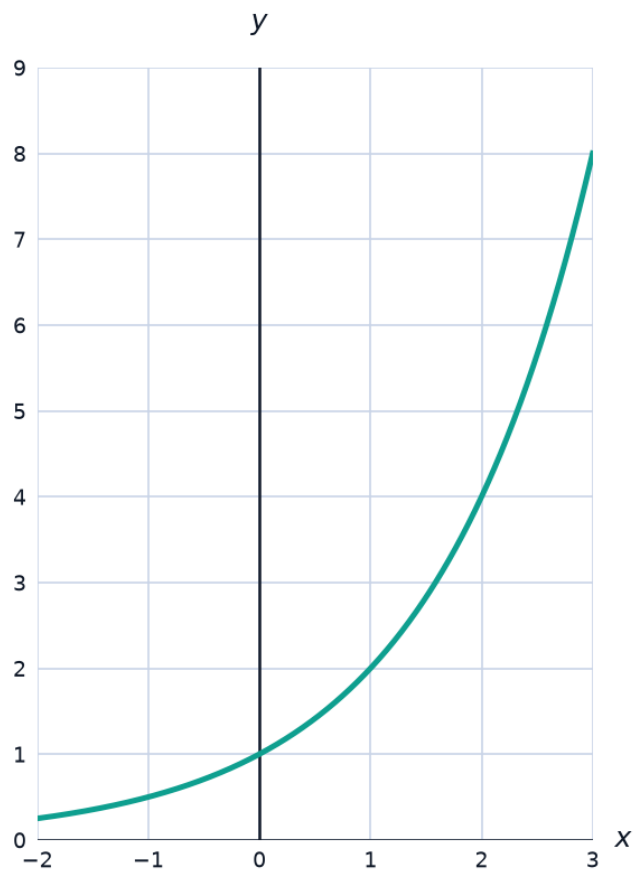
a. Neither

b. Cannot be determined

c. Decay

d. Growth

83 The graph shows an exponential function. Based on the graph, does it represent growth or decay?



a. Neither

b. Growth

c. Cannot be determined

d. Decay

MCQ — Word problems: exponential models in context

84 An investment of 3,000 dollars grows according to $A(t) = 3{,}000e^{0.05t}$ dollars, where t is in years. Find the investment's value after 8 years, rounded to the nearest dollar.

a. 4,200

b. 4,800

c. 4,475

d. 3,400

85 A cup of coffee cools according to $T(t) = 22 + 70e^{-0.08t}$ (°C), where t is in minutes. Find the temperature after 15 minutes, rounded to one decimal place.

a. 38.5°C

b. 50.0°C

c. 43.1°C

d. 22.0°C

86 A population of bacteria doubles every 3 hours, starting at 500. Using $P(t) = 500 \cdot 2^{t/3}$, find the population after 9 hours.

a. 4,000

b. 2,000

c. 8,000

d. 1,500

87 The value of a car depreciates according to $V(t) = 25{,}000(0.85)^t$ dollars, where t is in years. Find the car's value after 5 years, rounded to the nearest dollar.

a. 11,093

b. 12,750

c. 9,500

d. 10,625